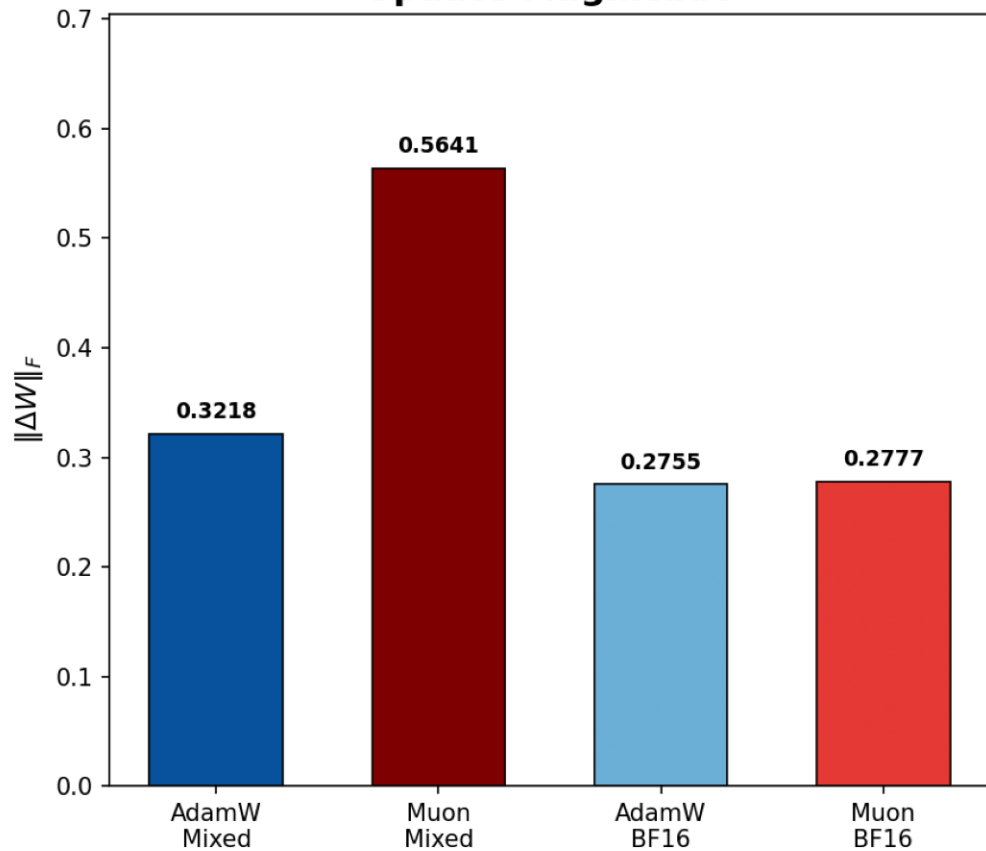
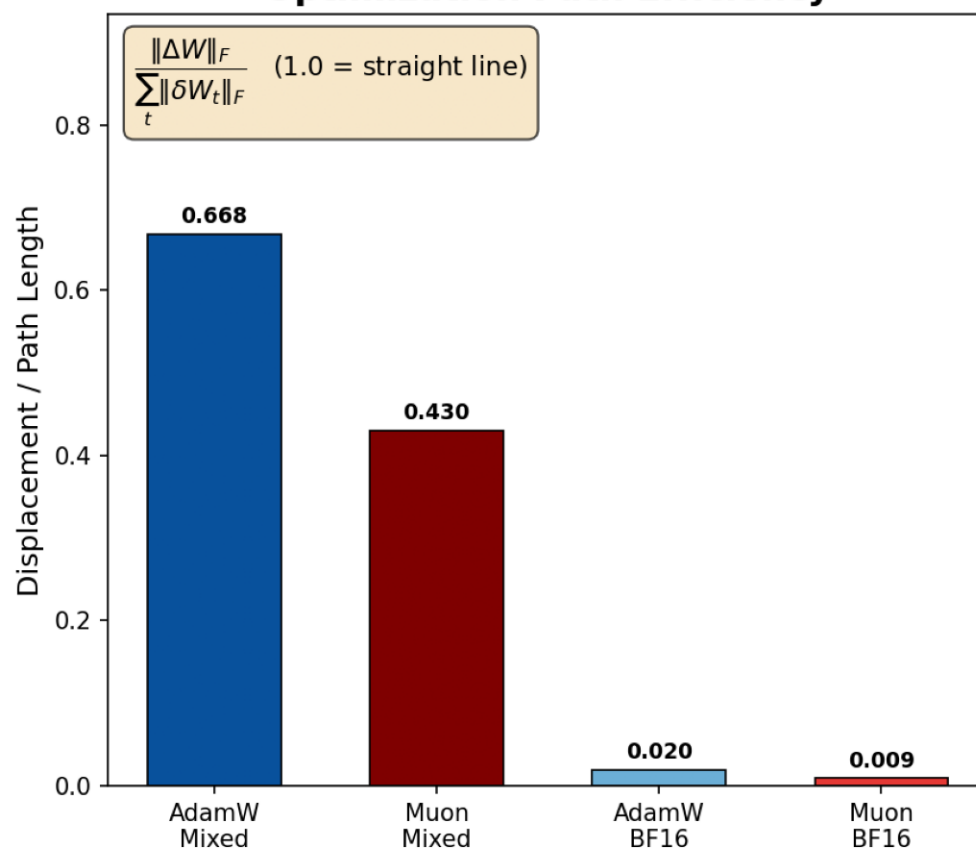


# BF16's near-zero path efficiency and 99%+ sparse updates are signatures of precision-induced truncation, not selective optimization.

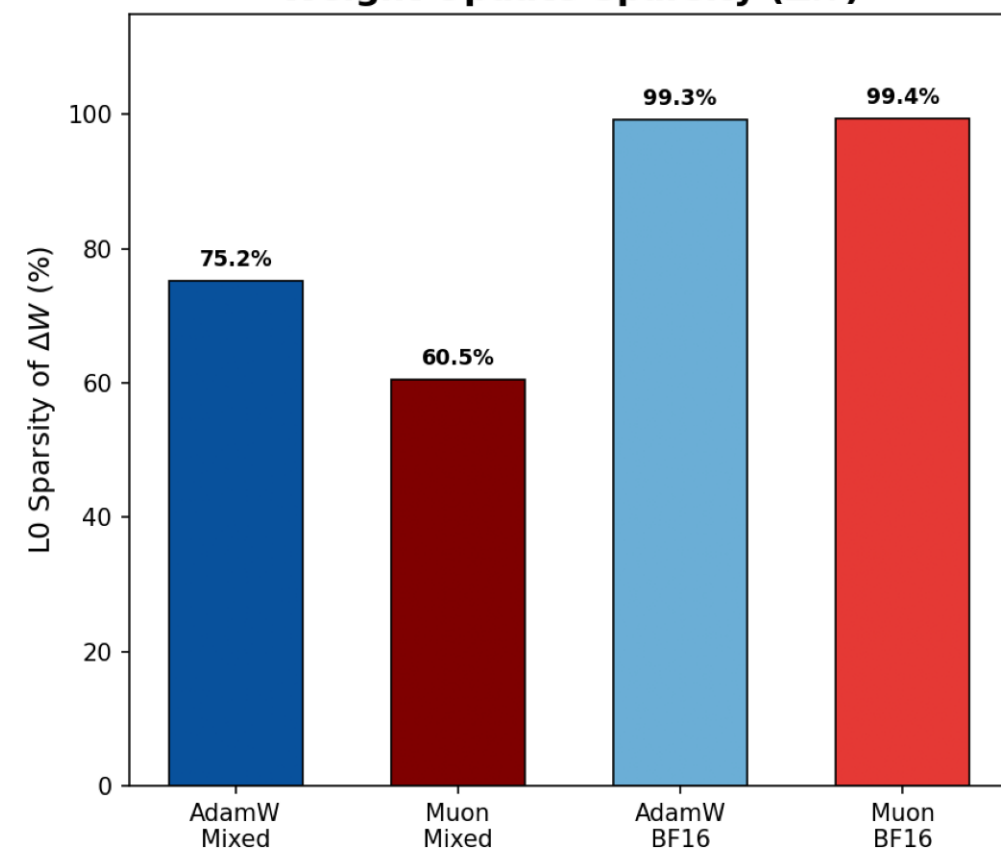
## Update Magnitude



## Optimization Path Efficiency



## Weight Update Sparsity ( $\Delta W$ )



*Evaluated at checkpoint with highest **validation accuracy** for each run.*